

Utrecht Summer School
Introduction to Complex Systems

Thursday project
Evolution in an agent-based model

1 Population dynamics with adaptive traits (*Python*)

In this final project, we will explore evolution due to natural selection in a simple model. The model falls in the category of agent-based models (ABMs), also called individual-based models (IBMs). These models explicitly simulate the dynamics of individual elements (agents) and their interactions.

Here, the agents are living creatures that live together as a population in a spatial domain. The environment offers a limited amount of food that the population depends on to survive and reproduce. If an individual has access to food, it will survive and, with a certain probability (the reproduction rate r), create offspring. If it does not have access to food, there is a chance (mortality d) it may die. Individuals can move around in the domain according to their *motility* m (quantifying how mobile an individual is).

The traits of each individual are thus encoded in its motility m , reproduction rate r and mortality d . The individual keeps these traits for its whole life and passes them on to its offspring. However, whenever new life is created, mutations are possible that slightly alter the offspring's traits. The question is: how will the distribution of these traits evolve over time in the population? Will some types be naturally selected compared to others?

To simulate this, we consider a two-dimensional domain with periodic boundary conditions (e.g., leaving the domain on one side brings you back on the opposite side). At every time step, we introduce a certain amount of food into the domain, distributed at random locations. Then, for every agent i at its current position (x_i, y_i) , we check whether it has access to food. If it doesn't, it dies with probability d_i . If it has access to food, it creates one child with probability r_i (if there are locally more individuals than food is available, the probability is reduced proportional to the food deficiency). Additionally, each living individual moves randomly over the domain within a maximal radius given by its motility m (mathematically, a direction between 0° and 360° and distance between 0 and m length units are chosen randomly).

During reproduction, mutations can happen for each trait m , d and r . The strength of the mutation (i.e. by how much the offspring differs from its parent) is controlled by parameters that set the maximally possible change for each trait. A mutation is then a random change that does not exceed this maximum. Furthermore, each trait has a possibility range, which by default is $m \in (0, 0.5)$, $d \in (0.1, 1.0)$ and $r \in (0.0, 1.0)$.

The agent-based model described above is implemented in Python as the `AgentEvolutionModel` class defined in `evolution_abm.py`. Use the Jupyter notebook `evolution.ipynb` to get started.

1.1 Dynamics without mutations

First, run several simulations (using the `simulate` function) without evolution, meaning that offspring will inherit exactly the parent's traits. This can be achieved by setting all maximum mutation parameters to zero. Analyze the evolution of food and agents. What do you observe? Play around a bit with the parameters m , d and r and see if and how the dynamics change. If you want, you can also change the food input over time by using a custom `food_input` function.

1.2 Limits of variation

Before turning on evolution, think about what values make sense for the traits m , d and r . Are the default ranges given above sensible? What can or cannot happen to an individual with these bounds?

1.3 Evolution

Once you have some intuition for the system without evolution, see how things change when the evolutionary parameters are non-zero. A good starting point would be to set the maximum variation in one of the parameters to something small like 0.01. How does the distribution of a trait evolve over time? Are certain values selected by evolution?

1.4 Additional suggestions

If time permits, continue exploring evolution in this model. For instance, you can study evolution under the variation of multiple parameters. Another possibility is to experiment with the food input function: what happens if the environment becomes harsher or more hospitable? Does this influence the evolution in the model? You can also make the food function dependent on the number of agents to incorporate a state-dependent food input.

Tip: The `simulate` function returns all simulation data, so you can analyze the dynamics in detail.

Lastly, do you think this model could be used in a different context than a population of creatures? For example, could it be interpreted as an economic or social community model? If so, think about what the individuals would represent, what the traits m , d and r would represent and what evolution would mean in that context.